

Family-Conditioned Multiview Occupancy Fitting for Semantic Proxy Actors in a Synthetic UAV Digital-Twin Setting (SPPA-MVFit)

Authors omitted for review

Abstract

Object-level UAV telemetry is not a mesh: a detector may provide a class family, confidence, track identifier, bounding region, and approximate pose, but no occupancy geometry. This paper studies whether a frozen semantic-family part graph improves lightweight 3D occupancy relative to an input-matched, equal-budget nonsemantic graph when both consume the same calibrated top and side silhouettes. The method, family-conditioned SPPA-MVFit, adjusts five shared parameters of a fixed eight-part primitive actor with a 31-candidate silhouette objective; text-only and silhouette-fitted modes call the same actor builder.

On a preregistered developer-held-out synthetic test of 240 actors balanced across six morphology families and two source strata (CSG-ID and structurally different implicit-OOD), the primary paired mean voxel-IoU difference of SPPA-MVFit minus Generic-MVFit is 0.190 with stratified bootstrap 95% CI [0.181, 0.199]. The lower bound exceeds the prespecified +0.030 superiority margin, so the primary hypothesis passes. Both strata remain positive (0.209 CSG-ID; 0.172 implicit-OOD). Secondary comparisons favour SPPA-MVFit over text-only SPPA and lightweight geometric baselines; median clean-call wall time is 9.4 ms (p95 10.6 ms) on the benchmark machine. The result is synthetic internal evidence only: it does not establish real UAV reconstruction, measured flight validity, operator benefit, or an image-to-3D SOTA ranking. A separate SPPA descriptor/update contract and optional Unreal Engine backend are reported as deployment context, not as the primary scientific endpoint.

Keywords: semantic 3D proxy; multiview silhouette fitting; family-conditioned part graph; occupancy IoU; synthetic spatial proxy; UAV digital-twin setting.

1 Introduction

Semantic-telemetry UAV digital twins replace part of the continuous video stream with object-level messages. A UAV-side perception stack can report object tracks, classes, confidence, positions, and approximate geometry. The ground station then has to represent those objects in an Unreal/Cesium digital twin. That architectural choice creates a rendering problem that is separate from the flight system itself: object-level telemetry is not a mesh.

Curated 3D assets are useful when the class is known, the asset library is complete, and run-time loading is acceptable. Those assumptions are weak for field UAV operation. Long-tail labels, ambiguous detections, partial views, occlusion, motion blur, and changing object proportions make a fixed asset library brittle. Neural image-to-3D and text-to-3D methods address a different objective: visual asset generation. They can produce richer geometry, but they depend on usable crops or prompts, GPU inference, and mesh outputs that may be too dense for a portable flight laptop that is also running Unreal, telemetry, video, and tracking under a shared rendering budget.

SPPA proposes a lower-fidelity representation point. Instead of reconstructing the real mesh, it compiles a detection into a semantic part graph made from primitive boxes, cylinders, ellipsoids, cones, and tori. A vehicle becomes a cab, body/cargo region, wheels, and optional top details. A quadruped becomes a torso, head, and legs. A tree becomes a trunk and canopy volumes. Unknown or unsupported labels are not hallucinated into specific objects; they fall back to a generic uncertainty-aware display volume with explicit uncertainty metadata.

The scientific question is therefore not whether SPPA looks better than neural 3D generation. It should not. The measurable question is whether a frozen semantic-family part graph improves lightweight 3D occupancy relative to an input-matched, equal-budget nonsemantic graph when both consume the same calibrated silhouettes. Runtime persistence and updateability are reported as separate engineering properties rather than collapsed into a utility score.

2 Contribution and Scope

The scientific contribution is **family-conditioned SPPA-MVFit**: evidence that a frozen semantic-family part graph improves 3D occupancy over an equal-budget nonsemantic eight-part graph under identical calibrated top/side silhouettes, parameter count, objective, bounds, initialization, and candidate budget. The knowledge representation under test is the family graph; the optimizer is deliberately simple and is not claimed as a novel fitting algorithm. The method is family-conditioned, not class-agnostic open-set fitting, and not universal image-to-3D reconstruction.

Secondary contributions provide deployment context only:

- **Shared generator contract.** Text-only and silhouette-fitted SPPA modes call the same actor builder and part graph; silhouette evidence changes only the five-parameter vector θ .
- **Runtime descriptor/update layer.** SPPA-DESC-0.2/ SPPA-UPD-0.2 separate create from pose/material/shape updates for optional Unreal consumption alongside curated assets.
- **Evidence boundary.** Confirmatory occupancy evidence is developer-held-out synthetic geometry. Real images in this package are qualitative probes without 3D ground truth. Operator benefit, live flight, dense VR readiness, and SOTA image-to-3D ranking remain outside the claim.

SPPA: language-assisted part decomposition compiled into deterministic 3D primitives

Language can draft the recipe offline; runtime compiles a reviewed versioned recipe plus telemetry.

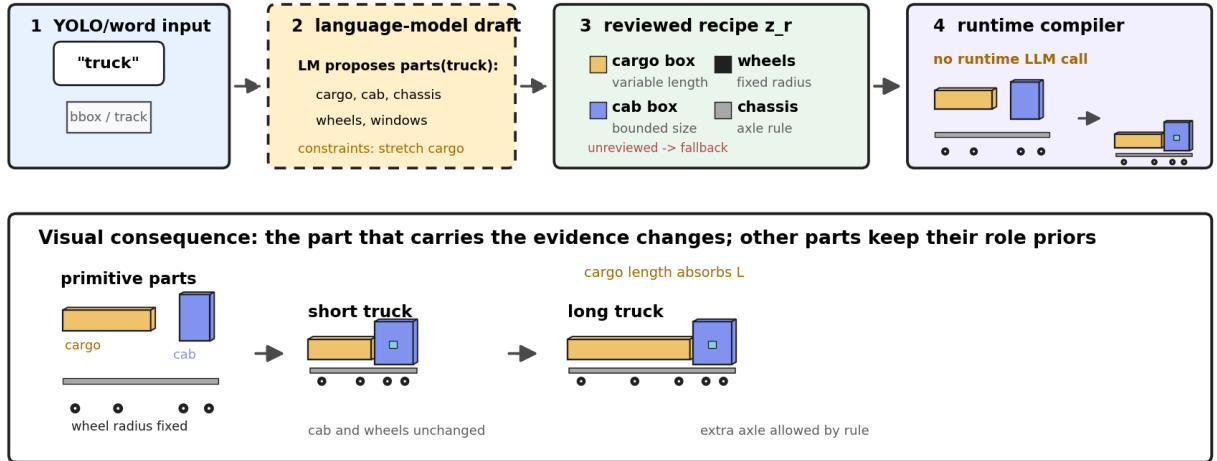


Figure 1: SPPA semantic-label-to-parts-to-3D mechanism for the truck archetype. A language model, ontology tool, or human author may draft candidate part roles offline, but runtime execution uses only a reviewed versioned recipe z_r . The figure shows the proposed chain explicitly: an input label or word activates an offline draft, the draft is reduced to reviewed part roles and bounded update rules, and runtime compiles those roles into primitive geometry. The short and long proxy variants illustrate the implemented role-specific rule: cargo length changes while cab and wheel priors remain bounded, so the object is not represented by root-scaling a single box.

Claim boundary: Figure 1 is not a claim of photorealistic reconstruction or universally correct geometry; SPPA is only claimed here as a bounded, reviewed semantic-proxy compiler.

Figure 1 states the central representation claim visually: SPPA is a reviewed semantic recipe compiler, not arbitrary word-to-3D generation and not a primitive-shape benchmark. The language component is limited to offline decomposition drafting and ontology expansion. The implemented runtime path starts after an admitted recipe exists: a detector label and telemetry state select a reviewed archetype, fill bounded part parameters, and expose explicit fallback behavior when the label is unsupported. For vehicles, the important distinction is part-parametric adaptation rather than global mesh or root-scale deformation.

3 Language Model Use and Geometry Path

The current SPPA implementation does not call a language model in the runtime path. The implemented resolver contract records `resolver_source` as a static keyword ontology and marks `runtime_llm_used` as false; the versioned recipe manifest also disallows runtime LLM use. The executable system described in this paper should therefore not be read as a live LLM that answers questions such as “is this a mammal?” or “how many legs should it have?” before constructing a mesh. Those facts are already encoded in reviewed archetype recipes. If an LLM is used at all, it is only an offline authoring aid that can propose candidate part decompositions; those candidates must be reviewed, versioned, regression tested, and admitted before runtime use. The present artifacts record that offline assistance category, but they do not record a specific LLM provider, model name, prompt, or output hash, so the paper makes no claim about a particular language model.

Two input modes are therefore distinct. A detector-driven input contains a class label plus confidence, track state, bounding region, optional mask, pose, and possibly calibrated scale. SPPA may use that evidence to choose a family, derive an axial footprint, update pose, or adapt dimensions. A tag-only input contains only a word or phrase; it can select a reviewed recipe and default dimension priors, but it cannot provide real silhouette, pose, color, or metric scale evidence. This distinction is important for the benchmark figures: a text-only prompt activates a recipe, whereas a detected object can carry measurement evidence.

Runtime execution is a deterministic compilation pipeline:

1. A detector label or tag is normalized to a SPPA tag and runtime label. For example, labels such as `electric pylon`, `agricultural vehicle`, or `vehicle` are mapped to reviewed semantic families such as vertical structure, farm vehicle, heavy vehicle, or generic vehicle when the evidence supports only that level.
2. The generator resolves the normalized label through exact-class rules, keyword-archetype rules, verifier-gated open-label proxy recipes, or explicit unknown fallback.
3. Metric dimensions come from supplied dimensions, calibrated footprint evidence, or bounded archetype priors. Without calibration, a mask remains image-space evidence and does not by itself become metric geometry.
4. The selected recipe emits role-labeled primitives, such as body, cab, cargo, wheel, torso, head, leg, trunk, canopy, mast, crossbar, or fallback volume. When parts are semantically attached, the recipe also emits explicit lightweight connector cylinders between contact points, for example bicycle frame tubes, rider-to-bike limbs, tower braces, or tractor-trailer hitches. Each primitive records type, local transform, scale, material role, and uncertainty metadata.
5. The descriptor is consumed by Unreal, which creates primitive components once per track and archetype and then applies pose, confidence, material, or compatible shape updates without rerunning a generative model.

For an animal-like object, SPPA does not infer anatomy from first principles or edit a mesh by hand at runtime. A label such as **cow**, **horse**, **dog**, or a conservative **quadruped** family selects a semantic morphology profile inside the same quadruped generator. Those profiles control torso proportions, head/neck placement, legs, tail, horn/h hoof hints, and optional markings, while the emitted proxy remains a low-polygon part graph rather than a watertight biological reconstruction. This explains why **biker** is visually cleaner from text alone: wheels and frame tubes are discrete geometric roles, whereas an animal is an organic shape represented only by bounded semantic priors unless image or metric evidence is available. A local text-only animal probe over **cow**, **bull**, **horse**, **dog**, **sheep**, **goat**, **deer**, and **quadruped** generated all cases through this shared profile path, with 836–1180 triangles and at most 0.61 ms in-process part synthesis before export. For a tractor or tractor-trailer, the system need not detect the exact taxonomic class to be useful: a detector output such as **vehicle**, **agricultural vehicle**, **truck**, or **trailer** can be normalized to a farm, heavy, or generic vehicle proxy. If the label is weak or adversarial, SPPA should prefer a conservative fallback volume over inventing a specific object. In the current implementation, a small set of useful UAV open-label cases such as **forklift**, **traffic cone**, **water tank**, **shipping container**, **drone**, and **hay bale** may pass through a deterministic recipe verifier before they are accepted as generated proxy recipes; otherwise fallback remains the runtime behavior.

This is also the honest novelty boundary. Primitive composition, CSG, shape programs, hierarchical semantic parts, visual-hull occupancy, procedural modeling, silhouette consistency, multi-view primitive optimization, and text-to-CAD are mature research areas [2, 38, 46, 28, 20, 59, 37, 29, 16, 17, 19, 36, 45, 11, 27]. Unlike data-driven hierarchical part discovery [28], SPPA freezes a small reviewed family graph and tests whether that fixed topology improves occupancy under a shared multiview silhouette budget. The scientific contribution is the measured family-conditioned occupancy gain (Section 9); the descriptor/Unreal runtime layer is deployment context only.

4 Representation Design Space

The relevant comparison is not a single “3D representation” baseline. A UAV digital twin can visualize a semantic detection through several families of representations, each making different assumptions about input evidence, runtime budget, geometry, and operator interpretation. Geometry-only displays such as boxes, ellipsoids, or billboards remain useful lower-bound controls, but they are not the scientific competitor. SPPA targets the gap between a generic box and a curated asset: a role-labeled primitive actor when only sparse semantic telemetry is available.

Table 1 states the resulting deployment policy. It deliberately gives priority to curated assets and mapping geometry when those are the correct tools.

Table 1: Operational representation policy for semantic detections.

Situation	Preferred representation	Reason	Boundary
Matched project asset exists	Curated asset backend	best appearance and lower measured cost in the cow/tower subset	asset library must match class and scale
Known semantic family, no asset	SPPA primitive actor	persistent, local, role-labeled 3D proxy from label/pose input	finite reviewed archetypes
Unknown or adversarial label	SPPA fallback volume	avoids inventing a specific object from weak semantics	not visually specific
Clean crop plus offline budget	Image-to-3D or asset authoring	richer mesh can be generated outside the flight update loop	no per-frame track contract
Planning or occupancy task	Occupancy / ESDF / map layer	represents navigable space and collision evidence	not a role-labeled display actor

5 Related Work

SPPA overlaps with several mature research areas, but it does not replace them. Text-to-3D and image-to-3D generation methods such as DreamFusion, Magic3D, Point-E/Shap-E, InstantMesh/Stable Fast 3D, and TRELLIS/Hunyuan3D-class systems target visual asset generation or reconstruction [34, 24, 30, 18, 44, 22, 54, 39, 3, 40, 41, 48, 49, 50, 25, 21, 53, 52, 58, 42, 15]. Their strength is fidelity when suitable visual or prompt evidence is available. SPPA targets a different runtime point: deterministic actor generation from sparse semantic telemetry, low polygon counts, and cheap per-frame updates. Recent benchmark work also highlights why a real image-to-3D ranking needs a shared reference set, metrics, and part/structure evaluation rather than a local figure assembled from heterogeneous inputs [43, 55]. For this reason, the artifact package includes an explicit SOTA-protocol readiness manifest and verifier. In the current artifact state that verifier keeps the comparative figure classified as an input-alignment audit, not as an image-to-3D SOTA leaderboard.

Monocular and open-vocabulary 3D detection estimate 3D boxes or object state from images [4, 56]; YOLOE supplies approximate open-prompt detection/segmentation evidence rather than universal recognition [47]. Animal pose and animal reconstruction methods estimate detailed skeletons or category-specific 3D shape [57, 23, 14]. These methods can provide stronger evidence to SPPA, but SPPA itself is a rendering/backend layer, not a detector or pose estimator.

Primitive representations have a long history. Marr’s volumetric primitives, recognition-by-components, superquadrics, learned part decomposition, hierarchical semantic parts, classical visual-hull occupancy from silhouettes, recent superquadric scene decomposition, residual primitive fitting, and positive/negative multiview primitives show that structured primitive geometry can encode compact shape abstractions [26, 2, 38, 33, 46, 28, 20, 10, 11, 27]. Neural primitive and CSG parsers show that networks can infer compact primitive or boolean programs from shape evidence [59, 37]. Procedural modeling, shape grammars, and learned shape programs show how compact programs can generate or abstract structured objects [29, 16, 17]. Text-to-CAD and point-cloud-to-code systems extend this program-generation line to natural-language CAD prompts and LLM-assisted CAD reconstruction [19, 36]. SPPA uses these ideas operationally. The scientific novelty tested here is the family-conditioned occupancy prior under equal-budget multiview fitting; the track-updateable descriptor/Unreal path is secondary engineering context, not the confirmatory endpoint.

Graphics systems already provide level-of-detail, impostors, billboards, streaming tiles, and virtualized geometry [5, 6, 9, 32]. 3D and dynamic scene graphs maintain semantic world structure [1, 35, 51]. Occupancy grids, OctoMap, ESDF, and TSDF maps support planning and mapping [13, 31]. SPPA should be read as a display actor representation that can consume such upstream evidence, not as a mapping substrate.

Finally, any operator-facing claim must be evaluated with human-factors methods. SAGAT, situation-awareness theory, and NASA-TLX are relevant to a future SPPA operator study [7, 8, 12]. The present paper contains no pilot study.

6 SPPA Contract

At time t , SPPA consumes an object state

$$x_t = \{i, \ell, c, b_t, m_t, \xi_t, s_t, \Sigma_t, h_t\},$$

where i is a track identifier, ℓ a detector label, c confidence, b_t a bounding region, m_t an optional mask or footprint, ξ_t a world pose estimate, s_t optional metric dimensions, Σ_t uncertainty, and h_t the track history. The resolver maps ℓ to one of three cases:

- `exact_class`: implemented label with a fixed archetype;

- **keyword_archetype**: out-of-template label mapped to a finite reviewed archetype, such as vehicle or vertical structure;
- **fallback_unknown_label**: unknown label rendered as a generic uncertainty-aware volume.

This is deliberately not universal word-to-3D generation. SPPA can accept an open detector tag at runtime, but the tag is normalized into a reviewed family or a conservative unknown fallback before geometry is emitted. A language model may assist offline ontology expansion, but runtime resolution must remain bounded, reviewed, cached, and able to reject over-specific mappings. Figure 1 makes this boundary explicit: language-model support belongs to offline archetype authoring or review, while the runtime path maps a detector label to a reviewed archetype, generic fallback volume, descriptor parts, and Unreal components. The SPPA observation path makes the metric-evidence boundary explicit through a **SPPA-OBS-0.1** observation record. It stores detector evidence, camera and flight geometry, a projected local footprint when available, a metric dimension proposal, and a covariance-style uncertainty envelope. Unless the record marks telemetry and metric ground truth as measured, the resulting pose and scale are treated as scenario-relative evidence for a proxy contract rather than as flight-localization truth. The current implementation includes a versioned archetype-recipe contract with 15 reviewed archetypes and 64 exact/keyword/fallback label checks. That contract verifies resolver status, allowed runtime archetype, required semantic material roles, and disabled runtime LLM use; it does not validate geometric correctness for arbitrary open-vocabulary labels. A separate offline recipe-review gate now tests five candidate mappings before caching: one accepted reviewed-family mapping, one fallback-only mapping, and three rejected mappings for visual artifact context, unreviewed topology/material roles, or attempted runtime LLM use. This is an implementation guard, not proof of open-vocabulary coverage.

6.1 Part Graph

Each archetype a defines a part graph

$$G_a = (V_a, E_a), \quad v_k = (p_k, r_k, \tau_k, \rho_k),$$

where p_k is a primitive type, r_k is its role, τ_k its transform relative to the actor root, and ρ_k role-specific parameters such as dimension priors, color/material role, and confidence tags. The current implementation uses boxes, cylinders, ellipsoids, cones, and tori.

Vehicle adaptation is the clearest implemented case. A long truck and a short truck should not be produced by globally scaling the same box. In the current vehicle grammar, cab and wheel dimensions remain bounded by role priors, while the cargo/body span and axle placement absorb length changes. In the synthetic truck check, two descriptors with equal width/height and different length keep cab scale fixed at $[1.656, 2.070, 1.952]$ m and wheel scale fixed at $[0.392, 0.094]$ m, while cargo/body length changes from 2.263 m to 5.188 m. This is an internal part-invariance diagnostic, not a visual comparison against competing 3D generation methods and not proof that SPPA recovers true truck morphology from UAV imagery. The corresponding component views are kept in the artifact log rather than used as main-paper comparative evidence.

6.2 Pose, Yaw, and Updates

The descriptor separates actor pose ξ_t from part parameters q_t . Pose changes are per-frame updates; topology regeneration is avoided unless the class/archetype changes or new evidence crosses a configured threshold. A typical update packet therefore changes only root transform, confidence, material/uncertainty tags, or a small set of shape parameters.

Yaw ambiguity is handled explicitly. A top-down footprint or PCA orientation gives an axial orientation modulo π , not necessarily a signed heading modulo 2π . Signed yaw requires velocity, asymmetry, multi-frame evidence, or an external heading source. If no reliable sign exists, the

descriptor sets `yaw_ambiguous=true`. The exact thresholds and weights in the fitting objective are exposed as versioned protocol parameters in the current implementation, not calibrated constants.

When a mask polygon is available, the descriptor records both the axis-aligned mask box and an oriented image-space footprint from PCA projections. This is used as footprint evidence and axial yaw evidence. If explicit metric dimensions or an explicit image-to-ground scale are supplied, the current generator can convert the calibrated footprint to metric length/width and pass those dimensions to the part-parametric builder. Without such calibration, the mask remains image-space evidence only and does not change metric geometry.

7 Implementation

The prototype has three layers. The first is a Python OBJ/MTL prototype generator for a finite set of explicit and keyword-resolved archetypes, including vehicles, vulnerable road users, vegetation, vertical structures, generic quadrupeds, and unknown fallback volumes. This path is useful for visual inspection and lightweight local timing only.

The second layer emits versioned descriptors and update packets. A `SPPA-DESC-0.2` record stores resolver source, match type, fallback reason, scale source, axis-aligned and oriented mask footprint evidence, yaw ambiguity, shape confidence, material source, optional observed-color evidence, part list, primitive counts, and scheduler metadata. A `SPPA-UPD-0.2` record separates create/regenerate payloads from pose/no-op/shape updates. The design goal is to generate once per track/archetype, update pose/material/uncertainty every frame, and apply shape-parameter changes in place when the part topology is unchanged. When supplied by the SPPA observation bridge, the descriptor additionally preserves the observation identifier, covariance matrix, position/scale/yaw uncertainty estimates, and visual uncertainty policy. These fields are carried as auditable metadata; they do not make the visual uncertainty encoding a validated human-factors result. In the current generator, `descriptor_id` is a topology-cache identity: it excludes pose, timestamp, confidence, and observed color, which remain in `input_hash` or update fields. This separation is required for cheap per-frame updates; otherwise a pose update would look like a new mesh descriptor.

The third layer is an optional Unreal Engine 5.7 telemetry backend. The same obstacle payload can drive either curated asset actors or semantic proxy actors. The asset backend remains the default. The SPPA backend can ingest descriptor JSON, construct primitive components from the `parts` array, apply pose/no-op/shape update packets, and reject invalid JSON or mismatched descriptor identifiers without changing the current actor. This implements a runtime switch without forcing a migration away from curated assets.

The current resolver coverage regression checks 34 direct labels, including reviewed archetypes, conservative fallback contexts, and verifier-gated open-label recipes. A separate recipe-manifest contract checks 23 archetypes and 95 exact or keyword labels against the runtime generator. These checks verify resolver consistency and recipe roles; they are not evidence that the ontology is complete.

8 Family-Conditioned SPPA-MVFit

8.1 Observation contract and shared actor

Every compared method receives the same case identifier, morphology-family token, metric world box, and two 96×96 orthographic binary silhouettes (top and side). No method receives RGB pixels, detector scores, source parameters, ground-truth occupancy, evaluation projections, or real or replay telemetry. The forbidden-input list is frozen in the protocol package.

An actor is an ordered list of eight primitive slots (boxes, cylinders, ellipsoids). Family graphs and a single `generic` graph are stored as static JSON. The only builder used by text-only SPPA,

Generic-MVFit, and SPPA-MVFit is

$$A = \text{Build}(g, \theta), \quad \theta = (\log s_x, \log s_y, \log s_z, s_{\text{sec}}, o_{\text{sec}}), \quad (1)$$

with identical bounds for every graph g . Text-only SPPA uses the family graph at the default θ_0 . Generic-MVFit and SPPA-MVFit optimize the same five parameters with the same silhouette disagreement objective and a fixed 31-candidate local search budget; only the graph identity differs (**generic** vs family). Baselines include axis-aligned box, ellipsoid, capsule, billboard, and nonsemantic visual hull from the same masks.

8.2 Preregistered confirmatory protocol

The primary hypothesis compares clean-mask 64-cubed voxel IoU of SPPA-MVFit against Generic-MVFit. Analysis units are 240 source actors (20 per family in each of two strata). The confirmatory estimand is the equal-stratum mean paired difference. H1 passes only if the lower endpoint of a two-sided 95% stratified actor-bootstrap percentile interval (10,000 resamples, seed 77157) exceeds +0.030. Seeds were derived from a NIST Randomness Beacon pulse after a pre-test freeze and a written triple-role protocol release under Amendment 03. Method outputs were sealed and hashed before private ground-truth evaluation. Development results were used only to implement the frozen package and are not confirmatory.

Source actors come from two strata: CSG-ID unions of solids with family topology, and implicit-ODD occupancy from superelliptic extrusions, taper/twist, toroidal sections, and spline-distance tubes. Source and method modules do not import each other. The held-out set is developer-held-out synthetic geometry, not an externally blinded public UAV benchmark.

9 Primary Results

Spatial estimand. The spatial quantity under test is object-level occupancy of a lightweight proxy actor in a shared metric twin frame. Calibrated orthographic top and side silhouettes provide two projected occupancy constraints; 64-cubed voxel IoU is the spatial-quality metric against independent synthetic source occupancy. Family tokens are observed inputs that select a frozen part graph (a spatial prior over role-labeled volumes), not inferred open-set labels. This is a synthetic spatial-proxy evaluation for digital-twin display actors; it is not a cartographic map task and not a measured UAV sensing product.

Table 2 reports the sealed confirmatory decision. Table 3 reports clean-condition method means and wall-clock costs. Table 4 reports secondary paired differences of SPPA-MVFit against the remaining baselines using the same stratified bootstrap interval construction.

Table 2: Preregistered primary endpoint on the held-out synthetic test ($n=240$ actors). Provenance: **synthetic_geometry**. H1 passes only if the stratified bootstrap 95% lower bound exceeds +0.030.

Quantity	Value
Primary mean Δ IoU (SPPA-MVFit – Generic-MVFit)	0.190
Stratified bootstrap 95% CI	[0.181, 0.199]
Prespecified superiority margin	+0.030
H1 decision (lower bound > margin)	PASS
Actors (analysis units)	240
CSG-ID stratum mean Δ	0.209
Implicit-ODD stratum mean Δ	0.172
Resolution sensitivity 48/64/80	PASS

Table 3: Clean-mask method summary on the held-out test ($n=240$). Mean/median are 64-cubed voxel IoU. Timings are single-call wall-clock milliseconds on the benchmark machine and are not Unreal frame times.

Method	Mean IoU	Median IoU	Median ms	p95 ms
SPPA-MVFit	0.557	0.563	9.43	10.60
Generic-MVFit	0.367	0.390	11.34	11.81
SPPA text-only	0.427	0.441	0.77	1.81
Visual hull	0.522	0.567	0.22	0.24
Ellipsoid	0.342	0.292	0.50	0.55
Capsule	0.325	0.260	0.95	1.07
Axis-aligned box	0.248	0.190	0.23	0.27
Billboard	0.173	0.148	0.23	0.26

Table 4: Paired clean-mask mean IoU differences of SPPA-MVFit minus each comparator (stratified actor bootstrap 95% intervals). Visual hull is a high-complexity geometry reference, not a lightweight runtime actor.

Comparator	Mean Δ IoU	95% CI low	95% CI high
Generic-MVFit (H1)	0.190	0.181	0.199
SPPA text-only	0.130	0.116	0.144
Visual hull	0.036	0.027	0.044
Ellipsoid	0.216	0.207	0.224
Capsule	0.233	0.224	0.241
Axis-aligned box	0.310	0.301	0.318
Billboard	0.384	0.375	0.393

H1 passes: mean Δ IoU = 0.190 with 95% CI [0.181, 0.199]. The confirmatory estimand is the equal-stratum paired mean difference; Table 3 is descriptive. The CSG-ID stratum mean difference is 0.209 and the implicit-OOD stratum mean difference is 0.172; neither is non-positive. Table 5 reports all twelve family $\times$ stratum cells from the sealed analysis: every cell is positive, with the smallest gain on compact vehicles under implicit-OOD ($\Delta = 0.043$) and the largest on rider-cycle under CSG-ID ($\Delta = 0.458$). Figure 2 shows illustrative held-out occupancy for three families.

SPPA-MVFit exceeds SPPA text-only (mean $\Delta = 0.130$, CI [0.117, 0.143]) and lightweight geometric baselines. Mean IoU of SPPA-MVFit (0.557) is slightly above the classical nonsemantic visual hull (0.522) [20], but visual hull is denser free occupancy without an eight-part role-labeled actor contract; it is a high-complexity geometry reference, not a defeated runtime competitor. Secondary comparisons use the same stratified bootstrap intervals; null-centered bootstrap p -values with Holm adjustment are recorded in the analysis artifact (Amendment 04). Single-pass clean-call wall time for SPPA-MVFit during the sealed evaluation is median 9.4 ms (p95 10.6 ms) on the benchmark machine; this is not a warm-timing Unreal frame-time claim and is not the protocol H4 multi-call cost design. A resolution-sensitivity check at grids 48/64/80 on the prespecified actor subset passes the protocol threshold. Candidate silhouette renders during fitting used the executed 96×96 observation grid on both arms (Amendment 04 alignment of protocol text with frozen code).

Table 5: Sealed family $\times$ stratum clean-mask mean Δ IoU (SPPA-MVFit – Generic-MVFit). All twelve cells are positive.

Family	CSG-ID Δ	Implicit-ODD Δ	Mean Δ
articulated_vehicle	0.113	0.125	0.119
branching_vertical	0.200	0.210	0.205
compact_vehicle	0.101	0.043	0.072
lattice_tower	0.175	0.163	0.169
quadruped	0.205	0.189	0.197
rider_cycle	0.458	0.299	0.379
Equal-stratum aggregate	0.209	0.172	0.190

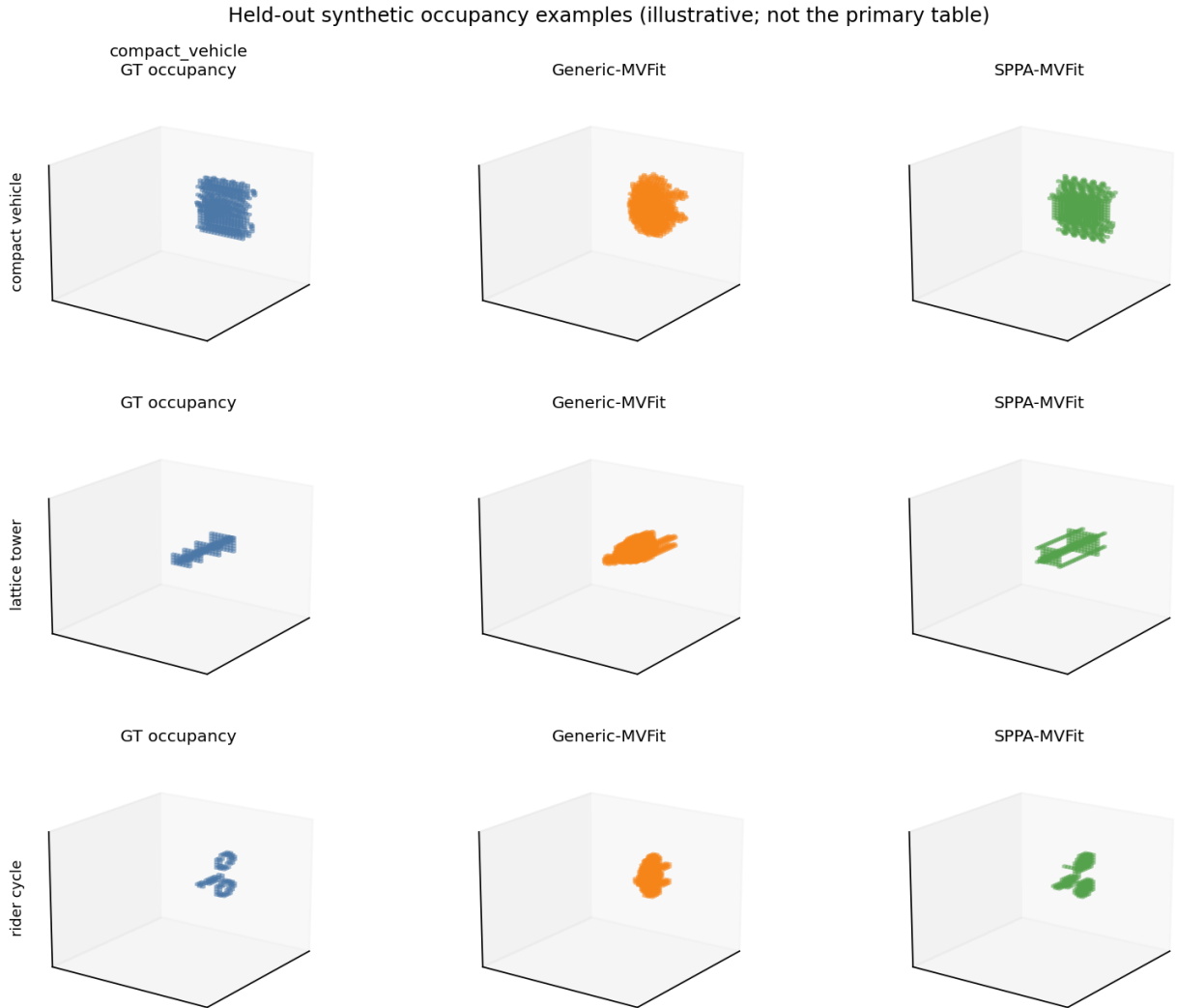


Figure 2: Illustrative synthetic occupancy for three held-out morphology cases (ground-truth source, Generic-MVFit, SPPA-MVFit). The figure is qualitative support for the tables; the confirmatory decision is Table 2.

Claim boundary: these occupancy gains are measured on synthetic calibrated silhouettes with known family tokens. They are not UAV flight measurements, not detector accuracy, and not operator-facing benefit.

10 Deployment Context (Secondary)

Outside the confirmatory endpoint, SPPA also exposes a versioned descriptor/update contract and an optional Unreal Engine backend that can coexist with curated assets on the same obstacle payload. Local engineering regressions support open-label recipe gating, connector primitives, and limited packaged desktop actor/HISM coexistence. Four user-supplied real images are used only as qualitative detector probes under declared replay assumptions; they have no 3D ground truth and do not support real-UAV reconstruction accuracy, SOTA image-to-3D ranking, or measured flight claims. Dense VR readiness, operator benefit, and radio-link advantage remain untested. Full historical systems tables live in the artifact archive, not in this submission body.

11 Discussion

The confirmatory result answers the preregistered question: under identical calibrated silhouettes and optimizer budget, a frozen family part graph improves lightweight occupancy over a generic eight-part graph by a large and stable margin. The knowledge representation, not a clever optimizer, is the active ingredient. Secondary gains over text-only SPPA show that silhouette evidence helps when the same family graph is held fixed. Gains over boxes, capsules, and billboards are expected but still quantified under the shared protocol.

Several limits remain binding. The held-out set is synthetic and developer-held-out; positive OOD-stratum differences do not license claims about real UAV masks, detectors, or metric flight geometry. Family tokens are supplied, so the experiment is family-conditioned rather than open-set. Nonsemantic visual hull can approach SPPA-MVFit IoU because it is free to materialize denser occupancy without an eight-part role-labeled actor contract; runtime persistence, updateability, and semantic roles are separate properties. Neural image-to-3D generators remain the wrong competitor for sparse telemetry without crops, but they remain preferable for offline asset authoring when budget allows. Unreal/HISM packaged measurements constrain deployment cost but do not establish dense VR readiness, headset performance, or operator benefit. A clean proxy can still look more certain than the underlying evidence; visual uncertainty encoding is specified, not human-validated.

12 Threats to Validity

First, confirmatory geometry is synthetic. Even the implicit-OOD stratum is internally generated, so external public UAV validity is untested. Second, family tokens are observed inputs; the experiment does not evaluate open-set family discovery. Third, silhouettes are clean orthographic projections plus prespecified corruptions, not real detector masks. Fourth, secondary systems evidence (Unreal, YOLOE real probes, neural-generator grids) uses different contracts and must not be pooled into a single quality score. Fifth, timing is local CPU wall-clock for the fitter, not flight-laptop Unreal frame time. Sixth, secondary inference uses stratified bootstrap intervals; null-centered bootstrap p -values with Holm adjustment are recorded in the sealed analysis artifact after Amendment 04. Seventh, the pre-test protocol release used a written local triple-role audit under Amendment 03 after external review infrastructure failure; it is not external peer review.

13 Claim Boundaries

Supported:

- On the sealed developer-held-out synthetic test, family-conditioned SPPA-MVFit improves clean voxel IoU over equal-budget Generic-MVFit beyond the prespecified +0.030 margin.
- The same shared generator supports text-only and silhouette-fitted modes; silhouette fitting changes θ only.

- SPPA descriptors can be consumed by an optional Unreal backend that coexists with curated assets on the tested packaged/replay paths.

Not supported:

- Real UAV 3D reconstruction accuracy or measured flight telemetry.
- Universal class-agnostic open-set geometry from arbitrary words.
- Operator benefit, safety improvement, or dense VR readiness.
- Image-to-3D SOTA ranking against neural generators.

14 Conclusion

Family-conditioned SPPA-MVFit shows that a reviewed semantic part graph is a measurable occupancy prior for lightweight multiview proxy fitting. Under a preregistered equal-budget comparison, SPPA-MVFit improves clean 3D occupancy over Generic-MVFit by 0.190 absolute IoU (95% CI [0.181, 0.199]) on 240 held-out synthetic actors, with positive CSG-ID and implicit-OOD strata and millisecond-scale local cost. The deployment-facing SPPA contract remains a separate engineering layer for compiling imperfect semantic telemetry into updateable Unreal actors when curated assets are missing or mismatched.

The honest next steps are external public data with metric ground truth, detector-derived silhouettes, and—only if claimed—operator studies. Until then, SPPA-MVFit should be judged as synthetic confirmatory evidence for a family-conditioned knowledge representation in lightweight proxy fitting, not as a flight system or universal reconstructor.

Data and Code Availability

The sealed SPPA-MVFit reproducibility package, protocol amendments, NIST-bound seed manifest hashes, and confirmatory summaries are included under `reproducibility/sppa_mvfit/`. Large Unreal artifacts and historical engineering logs remain supporting material and are not required to recompute the primary H1 tables.

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